

JINNAWAT KONSANTAD

Industrial AI & Smart Manufacturing | Industrial IoT & Data Analytics

Python | SQL | C# Basics | ESP32/IoT Sensors | Firebase | Anomaly Detection | Predictive Maintenance

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PROFESSIONAL SUMMARY

Industrial Engineering graduate and current M.Eng student focused on applying data analytics, Industrial IoT, practical automation, and AI-assisted workflows to real operational problems. Built and tested an ESP32-based IoT monitoring prototype during an internship project at Woraburi Sukhumvit for air-conditioner performance and water-leak detection using temperature/humidity data, vibration signals, Firebase real-time data logging, and LINE alerting logic. Strong project background in sensor-data acquisition, manufacturing-data workflow design, Excel VBA inventory automation, preventive maintenance planning, MTBF/MTTR analysis, anomaly-detection concepts, predictive-maintenance concepts, and dashboard logic. This background supports Smart Factory work involving production-data visibility, machine-condition monitoring, early issue detection, and data-driven process optimization.

RELEVANT TECHNICAL FOCUS AREAS

Industrial IoT Data Acquisition Hands-on tested prototype using ESP32/NodeMCU-class devices, temperature/humidity sensing, vibration sensing, water-leak sensing, Wi-Fi data transfer, and server/cloud logging.	Smart Factory Analytics Fit Project logic maps directly to real-time monitoring, anomaly detection, predictive maintenance, event logging, dashboard status, and technician alert workflows.
Manufacturing + IE Foundation IE background in maintenance analytics, MTBF/MTTR, inventory control, Lean/DMAIC, queue simulation, process improvement, and operational documentation.	Software / Data Workflow Base Python, SQL, Excel VBA, Streamlit, FastAPI/SQLite concepts, Firebase, Git/Linux basics, C# basics, and Gen AI-assisted technical documentation.

TECHNICAL SKILL STACK

Programming & Data: Python, SQL, Excel VBA/Macros, data collection, data cleaning, data validation, automated calculation, dashboard logic

Industrial IoT / Sensor Data: ESP32/NodeMCU concepts, temperature/humidity sensors, vibration sensors, water-leak sensors, Wi-Fi data transmission, Firebase Realtime Database, LINE Notify alerting

AI / ML Readiness: Data preprocessing, threshold-based anomaly logic, model training/evaluation fundamentals, predictive-maintenance concepts, scikit-learn workflow concepts, developing TensorFlow/Keras/PyTorch basics

Manufacturing / IE: MTBF/MTTR, maintenance planning, Lean, RCA, Work Study, Inventory, Logistics, ABC Analysis, PDCA, DMAIC, Fishbone, Why-Why

Software Workflow: Streamlit, FastAPI concepts, SQLite concepts, C# basics, Git basics, Linux basics, technical documentation, CI-style quality checks

Gen AI Productivity: ChatGPT, Claude, Gemini, prompt design, task decomposition, output verification, AI-assisted documentation

SELECTED PROJECTS & TECHNICAL CASES

IoT System for A/C Monitoring and Water-Leak Detection | Internship Project - Woraburi Sukhumvit / Tested Prototype

- Built and tested an ESP32/NodeMCU-class IoT prototype to collect room temperature/humidity, A/C compressor vibration, and water-leak status for smart facility monitoring.
- Designed the pilot architecture for a hotel-room monitoring use case, including Wi-Fi data transmission, Firebase Realtime Database logging, and LINE Notify alerts for technician response.
- Defined rule-based anomaly conditions such as temperature above 30°C for 5-10 minutes, compressor vibration mismatch, and immediate water-leak detection with room-level alert context.
- Documented the concept as a scalable pilot plan for hotel operations; the project was tested as a prototype during the internship and was not expanded to all hotel rooms.

Tools: ESP32/NodeMCU, DHT22/DS18B20 concepts, vibration sensor, water-leak sensor, Firebase, LINE Notify, Wi-Fi, anomaly logic, dashboard/event logging

Smart Manufacturing Data Workflow Prototype | Prototype / Concept Study

- Designed an end-to-end workflow for manufacturing data collection, preprocessing, dashboard logic, anomaly-detection concepts, predictive-maintenance concepts, and AI-assisted reporting.
- Structured a practical data pipeline concept covering data input, cleaning logic, storage, API-style workflow, dashboard presentation, and factory reporting.
- Connected IE thinking with machine/production-data visibility, maintenance insight, and Smart Factory analytics use cases.

Tools: Python, FastAPI concepts, SQLite concepts, Streamlit, dashboard logic, manufacturing data workflow, AI-assisted reporting

Lathe Preventive Maintenance Optimization | Maintenance Management Academic Project

- Developed a preventive maintenance plan for a TURNER 460x1000 lathe using historical failure data, MTBF, MTTR, ABC analysis, and PDCA.
- Analyzed failure patterns to support reliability-based maintenance planning, task prioritization, spare-part readiness, and structured execution planning.
- Built a maintenance-management case that supports predictive-maintenance thinking and early detection of equipment/process issues.

Tools: Preventive Maintenance, MTBF/MTTR, ABC analysis, PDCA, spare-part planning, maintenance scheduling

ADDITIONAL PROJECT EXPERIENCE

Inventory Management System with Excel VBA | Industrial Engineering Capstone Project

- Developed an Excel VBA-based inventory workflow for raw material tracking, structured input forms, validation logic, automated calculations, and stock visibility.
- Addressed stock-record discrepancies, unrecorded withdrawals, over-purchasing risk, and inconsistent data entry through standardized workflow design.
- Built structured input and validation logic to improve data consistency and reduce manual recording errors in operational workflows.

Tools: Excel VBA, Macros, inventory management, data validation, automated calculation, stock visibility

ELYSIOAI Platform | AI SaaS Platform MVP / Workflow Development

- Worked on a Next.js-based AI generation platform MVP involving Firebase/Firestore, Stripe billing flow, cost audit, credit-hold logic, billing-gate checks, quality scripts, mobile shell support, and technical documentation.
- Supported structured development workflows using linting, type checking, route-contract tests, billing checks, smoke tests, and CI-style quality gates.
- Practiced AI-assisted development workflow through task decomposition, verification, documentation, and quality-check habits.

Tools: Next.js, React, TypeScript, Firebase/Firestore, Stripe, Jest, Capacitor, Codex-assisted workflow

Lean / Logistics Process Improvement Cases | Academic IE Projects

- Applied Lean, DMAIC, Fishbone, Why-Why, queue analysis, ROI, and breakeven thinking to evaluate inventory, layout, service-capacity, and process-improvement scenarios.
- Practiced translating operational problems into structured analysis, improvement actions, and decision-support logic.

Tools: Lean, DMAIC, Fishbone, Why-Why, simulation, queue analysis, ROI, breakeven analysis

EDUCATION & CERTIFICATIONS

Education	Certifications
Master of Engineering in Industrial Engineering Suranaree University of Technology Expected May 2027 Bachelor of Engineering in Industrial Engineering Suranaree University of Technology Graduated High School: Science-Mathematics Program	• Data Analytics Essentials - Cisco • Python Essentials 1 - Cisco • TPM - Alison • Six Sigma Black Belt: Certified QFD Specialist - AIGPE/Udemy • Safety Management System Standards - SkillLane • Design Thinking - SET/CEA • Pitching Technique - SET • Presentation Skill - CEA

PROFESSIONAL STRENGTHS & LANGUAGES

Working Style	Languages
Responsible, detail-oriented, self-directed, comfortable with technical documentation, open to feedback, and effective in cross-functional work with factory, machine-development, IT, and data teams.	Thai: Native English: Working/intermediate communication; comfortable reading technical documentation